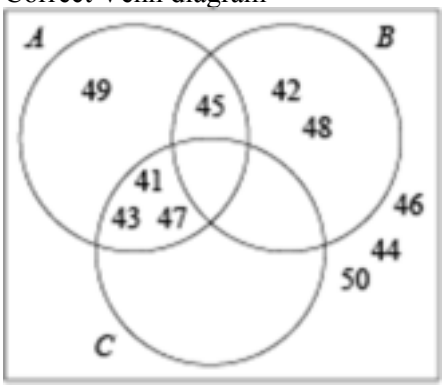
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 4(a) | Triangle at (–4, –4) (–1, –3) (–4, –3) | **2** | **B1** for correct points not joined or for reflection in any $y = k$ or for reflection in $x = –1$ |
| 4(b) | Triangle at (1, 1) (1, 4) (2, 4) | **2** | **B1** for correct points not joined or rotation 90 clockwise around any point or rotation 90 anticlockwise around (0, 0) |
| 4(c) | Translation $\begin{pmatrix} 5 \\ -6 \end{pmatrix}$ | **2** | **B1** for translation or correct vector oe |
| 5(a) | Correct Venn diagram   | **3** | **B2** for 8 or 9 numbers correct<br><br>or **B1** for 6 or 7 numbers correct |
| 5(b)(i) | 41, 43, 47 | **1** | **FT** *their* Venn diagram |
| 5(b)(ii) | 44, 46, 49, 50 | **1** | **FT** *their* Venn diagram |
| 5(c) | 0 | **1** | **FT** *their* Venn diagram |
| 6(a) | $y \geqslant x$ oe | **1** | |
| 6(b) | $2.25x + 1.5y \leqslant 22.5$ oe | **M1** | |
| | One step shown to $3x + 2y \leqslant 30$ | **A1** | |
| 6(c) | $y = 10$ ruled | **1** | Broken line |
| | $3x + 2y = 30$ ruled | **B2** | Solid line<br>**B1** for line passing through (0, 15) or (10, 0) |
| | $y = x$ ruled | **B1** | Solid line |
| | Correct region indicated | **B1** | |
| 6(d) | 412 | **2** | **M1** for (4, 9) identified or for evaluation $40x + 28y$ for an integer point in the region ($x > 0$ and $y > 0$) |
| 7(a) | $[BC^2 =]\ 80^2 + 115^2 – 2 \times 80 \times 115 \cos 72$ oe | **M1** | |
| | 118.06… | **A2** | **A1** for 13939… |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(b) | 67.8 or 67.9 or 67.83 to 67.88 | **3** | **M2** for [$\sin B =$] $\frac{115\times\sin 72}{118.1}$ oe<br>or **M1** for $\frac{115}{\sin B}=\frac{118.1}{\sin 72}$ oe |
| 7(c)(i) | 255 | **3** | **B1** for bearing of *B* from *A* is 75 soi<br>**M1** for 180 + 75 oe |
| 7(c)(ii) | [00]7.2 | **2** | **M1** for *their* **(c)(i)** – *their* **(b)** –180 |
| 7(d) | 11.8 or 11.82 to 11.83 | **3** | **M1** for 115 ÷ 35 oe<br>**M1** for *their* speed in m/s × 60 × 60 ÷ 1000 |
| 7(e) | 76.1 or 76.08 to 76.09 | **3** | **M2** for $\frac{\text{distance}}{80}=\sin 72$ oe<br>or **M1** for distance required is perpendicular to *AC* soi |
| 8(a)(i) | Correct sketch | **2** | **B1** for correct shape but inaccurate |
| 8(a)(ii) | Rotational [symmetry] order 2<br>[centre] (180, 0) | **2** | **B1** for rotational [symmetry] |
| 8(b) | 48.6 or 48.59 to 48.60<br>and 131.4 or 131.40 to 131.41 | **3** | **B2** for 48.6 or 48.59 to 48.60 or 131.4 or 131.40 to 131.41<br>or **M1** for sin *x* = 0.75 or better<br><br>If 0 scored, **SC1** for two answers adding to 180 |
| 8(c)(i) | $(x + 5)^2 – 11$ | **2** | **M1** for $(x + 5)^2 + k$<br>or $(x + \textit{their } 5)^2 + 14 – (\textit{their } 5)^2$ or $a = 5$ |
| 8(c)(ii) | Sketch of U-shaped parabola with a minimum indicated at (–5, –11)<br>with no part of graph in 4th quadrant | **3** | **FT** *their* $(x + 5)^2 – 11$ provided in that form<br>**B1** for U shape curve<br>**B1FT** for turning point at (–5, *k*) or (*k*, –11) |
| 9(a) | 39[.0] or 39.03 to 39.04… | **3** | **M2** for $\frac{165}{360}\times 2\times\pi\times 8 + 16$<br>or **M1** for $\frac{165}{360}\times 2\times\pi\times 8$ |
| 9(b) | 2.71 or 2.708… | **4** | **M3** for $\sqrt{\frac{\frac{165}{360}[\times\pi]\times 8^2}{4[\times\pi]}}$ oe<br>or **M2** for $r^2 = \frac{\frac{165}{360}[\times\pi]\times 8^2}{4[\times\pi]}$ oe<br>or **M1** for $\frac{165}{360}\times\pi\times 8^2$ oe seen |